Monte Carlo Simulation of Neutron Transport: Revisiting the ENIAC Era with Modern Benchmarks

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Abstract

We implemented a Monte Carlo simulation of neutron transport for the *Godiva* bare highly-enriched-uranium critical assembly, tracking individual neutron histories through fission, scattering, and capture via a power-iteration estimator for the criticality multiplier k. With multi-speed cross sections from ENDF/B-VIII.0 and Watt-spectrum fission birth energies, the simulation yields $\hat{k} = 1.10 \pm 0.006$ against the benchmark value 1.000 ± 0.001, with the residual 10% deviation attributable to a simplified inelastic-kinematics treatment. Using this as a controlled platform, we ask whether the choice of pseudo-random number generator (PRNG) measurably affects the criticality estimate. We compare von Neumann's middle-square method (8-digit decimal, as on ENIAC, implemented with a fixed-block scheme for controlled seed-to-seed comparisons) against a modern linear congruential generator and the Mersenne Twister, across ten independent seeds at each of several values of N (the number of neutron histories per power-iteration cycle). LCG and Mersenne Twister agree within statistical uncertainty for all N tested, while middle-square exhibits a $\sim 10\%$ systematic bias and a run-to-run standard deviation that does not track the $1/\sqrt{N}$ scaling predicted by the central limit theorem — staying flat with N at the von Neumann scale and behaving erratically at larger N. The variance-scaling failure is attributable to cyclic reuse of the fixed 2,000-number block; the bias is empirically robust but its mechanism is not isolated.

1 Introduction

1.1 Historical Context and Motivation

In the spring of 1948, a program ran on ENIAC — the first general-purpose electronic computer — that would quietly change the course of physics and computing alike. Stanislaw Ulam, recovering from illness and passing time playing solitaire, had grown fascinated by a deceptively simple question: rather than tracking the average behavior of a physical system through differential equations, what if one simply simulated the random trajectories of individual particles directly? He wrote to John von Neumann in 1947, and the two began work on what would become the Monte Carlo method [1].

The immediate application was *neutron transport* inside a fission weapon. Von Neumann and Richtmyer's 1947 internal report [5] framed the problem explicitly as a sampling procedure: track

individual neutrons probabilistically through fission, scattering, and absorption events, following each neutron and its descendants until the chain either sustains itself or dies out. Their preliminary estimate called for 100 primary neutrons through 100 collisions each, taking approximately 5 hours on ENIAC; the computation demonstrated that the method worked.

The stochasticity at the heart of Monte Carlo makes the method only as good as its source of pseudo-randomness, and von Neumann was acutely aware of this. He proposed and implemented the *middle-square* method on ENIAC; Hayes [7] documents that the recurrence cycle of the underlying iteration is highly seed-dependent and short for many seeds, so any practical use of the generator at ENIAC's operating scale was bounded above by the cycle length of the chosen seed — on the order of a few thousand distinct values, not the mathematical limit of the recurrence. A secondary focus of the present work is to revisit, with modern infrastructure, what consequences this constraint has for criticality estimates.

1.2 The Neutron Transport Equation

The governing equation of neutron transport is the *linearized Boltzmann transport equation*, a balance equation for the angular neutron flux $\psi(\mathbf{r}, E, \mathbf{\Omega}, t)$ — the density of neutrons at position $\mathbf{r} \in \mathbb{R}^3$, kinetic energy $E > 0$, direction $\mathbf{\Omega} \in S^2$, and time t:

$$\underbrace{\frac{1}{v}\frac{\partial \psi}{\partial t}}_{\text{rate of change}} = \underbrace{Q}_{\text{external source}} + \underbrace{\iint \Sigma_s \, \psi \, dE' \, d\Omega'}_{\text{scatter-in}} + \underbrace{\frac{\chi}{4\pi} \iint \nu \Sigma_f \, \psi \, dE' \, d\Omega'}_{\text{fission birth}} - \underbrace{(\mathbf{\Omega} \cdot \nabla)\psi}_{\text{streaming loss}} - \underbrace{\Sigma_t \, \psi}_{\text{collision loss}} \tag{1}$$

Each term has a direct physical reading. The streaming term records neutrons leaving an infinitesimal phase-space volume by simply moving through it; the collision term records neutrons removed by any interaction whatsoever, with Σ_t the total interaction probability per unit path length. The scatter integral reintroduces those that survive scattering, weighted by the appropriate *differential scattering cross section* Σ_s from incoming to outgoing energy and direction. The fission term creates $\nu \approx 2.58$ new neutrons per fission event, with their birth energies drawn from a known spectrum $\chi(E)$. The quantity v is the neutron's speed. The fully spelled-out equation, with all spatial and energy dependencies explicit, is reproduced as Eq. (8) in Appendix A.

The equation lives in a six-dimensional phase space plus time, and is analytically intractable in any realistic geometry. Two solution strategies exist. *Deterministic methods* discretize the phase space — energy groups, discrete ordinates for angles, spatial mesh — and solve the resulting linear system [3]. *Stochastic (Monte Carlo) methods* instead simulate individual neutron histories as random walks, whose ensemble average satisfies Eq. (1) in the limit of many samples [2]. Our work uses the latter.

1.3 Project Objectives

Two questions structure this project. First, how does the precision of the Monte Carlo estimate of k scale with the number N of neutron histories per power-iteration cycle? The expected answer is $1/\sqrt{N}$ from the central limit theorem applied to the per-cycle sample mean, but this prediction rests on assumptions of independence between draws that may or may not hold for a given pseudo-

random generator. Second, does the choice of PRNG — specifically, the historical middle-square method operated at its historical scale, against modern alternatives — introduce measurable bias or excess variance in the criticality estimate? Both questions are taken up in Sections 3–4.

2 Model and Methods

2.1 Geometry, Composition, and Modeling Assumptions

We simulate a bare, homogeneous sphere of highly enriched uranium (HEU) using the *Godiva* assembly (HEU-MET-FAST-001) as our benchmark. Godiva was assembled at Los Alamos in the 1950s and built to be exactly critical, meaning that the chain reaction is precisely self-sustaining ($k = 1$). More generally, $k < 1$ is subcritical (the chain reaction dies out), $k > 1$ is supercritical (it grows without bound), and $k = 1$ marks the boundary. The benchmark value $k_{\text{eff}} = 1.000 \pm 0.001$ [6] serves as our validation target. The key modeling choices are:

- **Geometry.** Perfect sphere of radius $R = 8.7407$ cm with vacuum boundary — neutrons reaching the surface escape and are removed from the population. All geometric and isotopic parameters are reproduced from Frankle [6].
- **Composition.** Homogeneous mixture of U-235 (93.77%), U-238 (5.20%), and U-234 (1.03%) by atom fraction, with total number density $n = 4.7984 \times 10^{-2}$ atoms/(barn·cm). Per isotope: $n_i = f_i\, n$.
- **Multi-speed transport.** Cross sections are evaluated at the neutron's current energy at every collision, using log-log interpolation of ENDF/B-VIII.0 [8] tabulated values. Birth energies for fission neutrons are sampled from the Watt spectrum (parameters for U-235; sampling algorithm in Appendix B).
- **Inelastic scattering treated with elastic kinematics.** Inelastic scattering — in which the neutron survives a collision but deposits energy into nuclear excited states — is included as a separate channel in the reaction sampling, but the post-collision energy is computed using the elastic kinematic formula rather than a proper inelastic-level treatment. This is the dominant remaining modeling limitation; we return to it in Section 4.
- **Constant fission yield.** The mean number of fission neutrons $\bar{\nu}$ is held fixed at the U-235 fast-region value 2.58 [2] regardless of incident energy; the rigorous Boltzmann equation (Appendix A) carries an explicit $\nu(E')$. This simplification is consistent with the inelastic choice above — neutrons in our model retain $\geq 98\%$ of their energy per scatter and so spend their entire lives in the 1.5–2.5 MeV band where $\bar{\nu} \approx 2.58$ is close to the true energy-dependent value, so constant $\bar{\nu}$ adds no separate error beyond what the inelastic treatment already imposes.
- **Markov property.** The future of a neutron depends only on its current state $(\mathbf{r}, \mathbf{\Omega}, E)$, not its past; no time-correlation effects are tracked. Reaction byproducts other than neutrons (gamma rays, fission fragments) are discarded.

- **Prompt only.** All fission neutrons are treated as prompt; delayed neutrons ($\sim 0.65\%$ for U-235) are neglected. This has no effect on the steady-state criticality estimate.

2.2 Cross Sections and Reaction Probabilities

Each neutron has a state $(\mathbf{r}, \mathbf{\Omega}, E)$. Between collisions it travels in a straight line. The probability of interacting in a path element dx is governed by the *macroscopic cross section*

$$\Sigma_t(E) = \sum_i n_i \, \sigma_t^{(i)}(E) \tag{2}$$

where $\sigma_t^{(i)}(E)$ is the microscopic total cross section of isotope i in barns ($1\,\text{barn} = 10^{-24}\,\text{cm}^2$). Throughout this report, lowercase σ denotes a microscopic (per-nucleus) cross section and uppercase Σ denotes the macroscopic counterpart (a per-unit-length reaction rate, summing over the homogeneous mixture); the superscript (i) tags the isotope (U-234, U-235, or U-238) and the subscript $x \in \{t, \text{el}, f, \gamma, \text{inel}\}$ tags the reaction channel (total, elastic, fission, radiative capture, inelastic). The barn is not a geometric quantity derivable from first principles: it is an effective interaction probability that must be measured experimentally and is tabulated as a function of energy in evaluated nuclear-data libraries such as ENDF/B-VIII.0 [8]. Each reaction channel has its own microscopic cross section $\sigma_x^{(i)}(E)$, and the corresponding macroscopic quantity is $\Sigma_x(E) = \sum_i n_i \sigma_x^{(i)}(E)$.

The exponential distribution of free-flight path lengths follows from treating collisions along a track as a Poisson process with rate Σ_t per unit length: divide the track into infinitesimal elements dx, each an independent Bernoulli trial with success probability $\Sigma_t \, dx$. The distance to the first collision is then $\text{Exp}(\Sigma_t)$ with mean free path $\lambda = 1/\Sigma_t$ [2]. Sampling formulas for path length, isotope, reaction type, and post-scatter energy/direction are given in Appendix C.

For orientation, Table 1 reports the macroscopic cross sections at $E = 2$ MeV — approximately the mean energy of the U-235 fission spectrum, used here only as a representative reference point. At runtime, all cross sections are recomputed per-collision from ENDF/B-VIII.0 interpolators.

Table 1: Macroscopic cross sections at $E = 2$ MeV for the Godiva HEU composition, evaluated from ENDF/B-VIII.0 [8]. Values are shown for orientation; the simulation interpolates per-collision. The inelastic channel is computed as the residual $\Sigma_t - \Sigma_{\text{el}} - \Sigma_f - \Sigma_\gamma$.

Reaction	Σ (cm^{-1})	Comment
Total	0.3480	governs path length
Elastic	0.1810	momentum transfer, neutron survives
Fission	0.0601	drives chain reaction
Capture (n, γ)	0.0028	neutron absorbed
Inelastic	0.1041	residual; treated as elastic kinematically
Mean free path	2.87 cm	$\lambda = 1/\Sigma_t$

2.3 Power Iteration for k-effective

The criticality multiplier k is estimated by power iteration over *cycles* (also called *generations*). Each cycle starts with N source neutrons sampled from the fission sites of the previous cycle. After all N histories are tracked, the resulting fission sites seed the next cycle (renormalized back to N neutrons by uniform resampling). The per-cycle estimate is

$$k_{\text{cycle}} = \frac{\text{fission neutrons produced in this cycle}}{\text{neutrons started in this cycle}}. \tag{3}$$

The first N_{inactive} cycles are discarded as the spatial distribution of source neutrons converges to its true shape (the fundamental mode of the transport operator) — the analogue of MCMC burn-in. The final estimate and its standard error are

$$\hat{k} = \frac{1}{N_{\text{active}}} \sum_{c=1}^{N_{\text{active}}} k_{\text{cycle},c}, \qquad \sigma_{\hat{k}} = \frac{\mathrm{std}(k_{\text{cycle},c})}{\sqrt{N_{\text{active}}}}. \tag{4}$$

The $1/\sqrt{N_{\text{active}}}$ scaling of $\sigma_{\hat{k}}$ is the central limit theorem applied to the sample mean of (approximately) i.i.d. per-cycle estimates. An analogous $1/\sqrt{N}$ scaling holds for $\mathrm{std}(k_{\text{cycle}})$ as N grows, since each k_{cycle} is itself a sample mean over N neutron histories. The combined prediction is therefore $\sigma_{\hat{k}} \propto 1/\sqrt{N \cdot N_{\text{active}}}$, and any empirical violation of the scaling at fixed N_{active} indicates that the i.i.d. assumption is breaking down — the central finding of Section 3.

For all studies in this work we use $N_{\text{inactive}} = 20$ inactive cycles, more than sufficient for source-shape convergence in a bare critical sphere — empirically, the spatial source distribution stabilizes within $\sim$10 cycles, consistent with the dominance ratios tabulated for benchmark assemblies of this size [4].

2.4 Pseudo-Random Number Generators

We implement three generators of progressively higher statistical quality, the first historically authentic and the latter two used extensively in modern Monte Carlo work.

Middle-square method (Hayes 2022 [7]). Squaring an n-digit number and extracting the middle n digits. Critically, von Neumann's implementation on ENIAC was *decimal*, not binary: 8 decimal digits, with the recurrence

$$x_{i+1} = \left\lfloor \frac{x_i^2 \bmod 10^{12}}{10^4} \right\rfloor \tag{5}$$

producing values in $[0, 10^8)$ (which we normalize to $[0, 1)$). The *recurrence cycle* — the number of iterations before Eq. (5) returns to a previously visited state — is highly seed-dependent; for 8-digit decimal seeds the median is approximately $2{,}700$, but the distribution is heavy-tailed and many seeds degenerate to 0 or short repeating sequences within a few hundred iterations [7]. In our implementation we adopt a fixed-block scheme to make seed-to-seed comparisons controlled: at construction we generate `block_size` $= 2000$ numbers from the seed, then index modularly on every call. This makes the effective period the block size rather than the recurrence cycle of the originating seed (which may be longer or shorter), and gives a uniform basis for comparing across

seeds. We use the term *cycle* elsewhere in this report (§2, §3) to mean a power-iteration cycle in the simulation; the recurrence cycle of the PRNG is referenced only here and in the limitations of §4.

Linear Congruential Generator (LCG). The recurrence $x_{i+1} = (ax_i + c) \bmod m$ with Numerical Recipes parameters $(a, c, m) = (1{,}664{,}525,\ 1{,}013{,}904{,}223,\ 2^{32})$, which satisfy the Hull-Dobell conditions [9] for a full-period generator and therefore have period $m = 4 \times 10^9$.

Mersenne Twister (MT19937) [10]. Period $2^{19937} - 1$, passes all standard statistical randomness tests, and is the default generator in Python's `random` module. We use it as the reference "high quality" RNG.

A first cheap diagnostic is the consecutive-pair scatter (x_i, x_{i+1}) for $i = 1, \ldots, n$, shown in Fig. 1 for all three generators. All three panels show samples that look approximately uniformly distributed on $[0, 1]^2$ — importantly, including middle-square. A subtler feature is visible on inspection: the middle-square panel appears *less dense* than the other two even though the same total number of samples is plotted. This is because successive draws cycle through the same $\sim$2,000 values, so the same points are overplotted multiple times. The total number of *distinct* points is bounded by the block size, while LCG and Mersenne produce 10,000 distinct points. That this overlap is invisible at the level of marginal uniformity but dominates the simulation behavior is the core observation of Section 3.3.

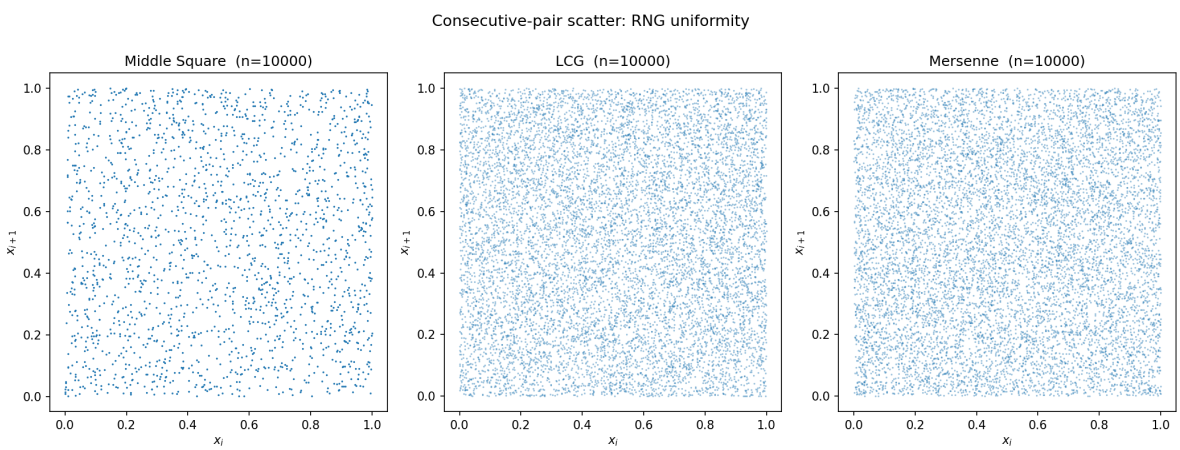

Figure 1: Consecutive-pair scatter (x_i, x_{i+1}) for $n = 10{,}000$ draws from each of the three PRNGs. All three pass the test of marginal uniformity. The middle-square panel is visibly less dense than the other two: the same $\sim$2,000 values are plotted on top of each other roughly five times due to the cyclic block reuse, but each pair (x_i, x_{i+1}) is also reused, so the overplot is exact rather than a thickening of independent points. This visual artifact will not appear in any single static frame; nevertheless, *the same correlations dominate the simulation behavior*, as Section 3.3 shows.

2.5 Multi-Seed Evaluation Methodology

The standard MC error estimator $\sigma_{\hat{k}}$ defined above captures *within-run* variability: it tells us how scattered $k_{\text{cycle},c}$ is across the N_{active} active cycles of one simulation. This is the right diagnostic when neutron histories are independent draws, but it is silent on a different source of variability:

how much would the estimate change if the same simulation were re-run from a *different RNG seed*?

To answer that question we adopt a multi-seed protocol. For each (RNG, N) combination, we run n_{runs} independent simulations, each seeded by a different value, and report the across-run statistics

$$\bar{k}_{\text{seeds}} = \frac{1}{n_{\text{runs}}} \sum_{r=1}^{n_{\text{runs}}} \hat{k}_r, \qquad \sigma_{\text{seeds}} = \text{std}(\hat{k}_r). \tag{6}$$

For all seed-dependent runs we use $n_{\text{runs}} = 10$ at von Neumann scale ($N \in \{50, 100, 150, 200\}$) and at larger N ($N \in \{200, 500, 1000\}$).

The two error measures $\sigma_{\hat{k}}$ and σ_{seeds} should agree if and only if cycle-to-cycle variability fully captures the simulation's stochasticity. Where they disagree, the disagreement is itself a finding: it means the RNG is contributing seed-to-seed variability beyond what within-run sampling explains.

3 Results

3.1 Validation Against the Godiva Benchmark

The primary validation run uses the Mersenne Twister at $N = 1{,}000$ histories per cycle, $N_{\text{inactive}} = 20$, $N_{\text{active}} = 50$, yielding

$$\hat{k} = 1.1008 \pm 0.0055 \tag{7}$$

against the benchmark value $k_{\text{eff}} = 1.000 \pm 0.001$ [6]. The reported uncertainty is the within-run standard error; across-seed variation at this N is somewhat larger ($\sigma_{\text{seeds}} \approx 0.014$, see Table 4) but does not change the conclusion. The simulation overpredicts k by 10.08%, well outside the statistical uncertainty band of either the simulation (± 0.005) or the benchmark (± 0.001). We attribute this deviation to the inelastic-as-elastic kinematics treatment described in Section 2: physical inelastic scattering on U-235 deposits $\sim$0.5–1 MeV per event into nuclear excited states, softening the neutron spectrum, while the elastic kinematic formula applied with mass number $A = 235$ produces at most 1.7% energy loss per scatter ($\alpha = ((A-1)/(A+1))^2 \approx 0.983$; mean loss $\sim 0.85\%$). Over a typical neutron history of 5–10 scatters, neutrons remain within $\sim$5–10$\%$ of their birth energy — well above the energies where physical inelastic scattering would have softened the spectrum. Neutrons in our simulation therefore stay near their Watt birth energies of $\sim$2 MeV throughout their lives, where $\bar{\nu} \approx 2.58$ rather than the ~ 2.45 a properly softened spectrum would sample, a 3–5$\%$ effect that propagates directly to $\hat{k}$. This $\bar{\nu}$ effect contributes to the overshoot; the larger mechanism is the rise of U-235's capture-to-fission ratio σ_c/σ_f as the spectrum softens (from ~ 0.05 at 2 MeV to ~ 0.3 at 100 keV) — a softened spectrum loses neutrons to capture more readily relative to fission, pushing k down.

3.2 Spatial Fission Source Profile

The power-iteration scheme has an interpretation we have not yet exploited. The $N_{\text{inactive}} = 20$ burn-in cycles serve to relax the source distribution from its initial delta function (all neutrons at the origin) to the dominant eigenmode of the transport operator, which is the steady-state *fission source*

density $S(\mathbf{r}) \propto \int \nu\Sigma_f(E)\,\phi(\mathbf{r}, E)\,dE$ — a close proxy for the spatial flux profile. After burn-in, the fission positions emerging from each active cycle are samples from this converged distribution.

Fig. 2 plots the radial fission source density, computed by histogramming all fission positions accumulated across the 50 active cycles of the validation run (n $= 55{,}038$ positions), binning by radius, and dividing each bin's count by its shell volume $4\pi r^2 \Delta r$. For comparison we overlay the diffusion-theory fundamental mode $S(r) \propto \sin(\pi r/\tilde{R})/r$ with $\tilde{R} = R + 0.7104\lambda_{\text{tr}} \approx 10.8$ cm the extrapolated radius (Milne's transport result; see e.g. Bell & Glasstone [11]).

The simulation is qualitatively consistent with the expected center-peaked profile. The apparent dip at $r = 0.22$ cm is an artifact of bin-1 statistics: only 14 of the $55{,}038$ fission positions land in that innermost bin (its shell volume is 0.35 cm^3, the smallest of the 20 bins), and with so few events the density estimate is statistically unreliable. At larger radii the simulation tracks the diffusion-theory curve in shape and falls modestly below it: the peak-normalized gap reaches ~ 0.20 across the mid-region ($r \approx 2$–5 cm) and narrows again toward the boundary. The residual offset is consistent with the simple diffusion approximation neglecting higher-order transport corrections; that the gap is small and roughly uniform across the bulk of the sphere indicates that the corrected extrapolation length captures most of the boundary effect, leaving a leading-order diffusion underestimate of the angular flux in the interior. The match in shape visualizes what the burn-in cycles accomplished, taking the source from a delta function at the origin to a center-weighted distribution that fills the sphere with the expected fundamental-mode profile.

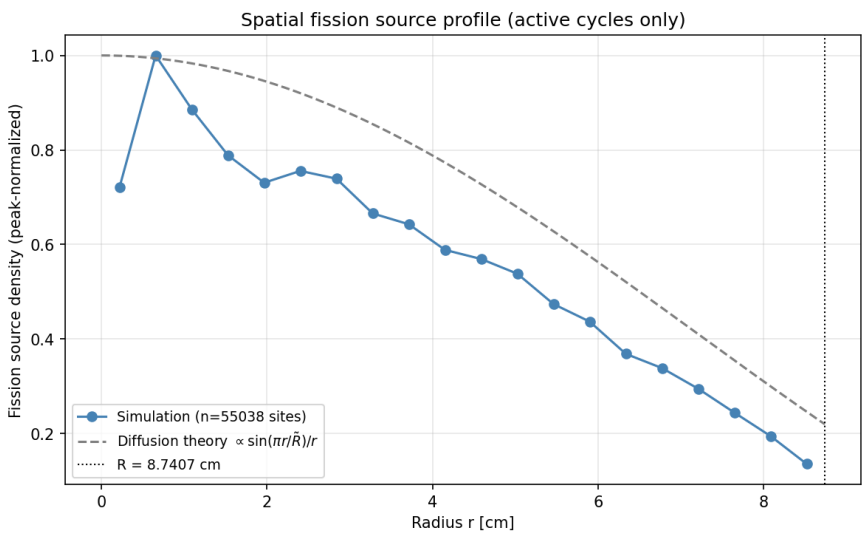

Figure 2: Spatial fission source density (peak-normalized) computed from $55{,}038$ fission positions accumulated over the 50 active cycles of the validation run (Mersenne Twister, $N = 1{,}000$). The dashed grey curve is the diffusion-theory fundamental mode $\sin(\pi r/\tilde{R})/r$ with $\tilde{R} = R + 0.7104\lambda_{\text{tr}}$. The simulation tracks the diffusion-theory shape but runs modestly below it across the bulk of the sphere (peak-normalized gap ~ 0.20 around $r = 2$–5 cm, narrowing toward the boundary), consistent with leading-order corrections that simple diffusion theory does not capture.

3.3 PRNG Comparison at von Neumann Scale

The central question of this study is whether the choice of PRNG measurably affects the criticality estimate at the historical operating point of the 1948 ENIAC simulations ($N \approx 100$ histories per cycle). Using the multi-seed protocol of Section 2.5, with $n_{\text{runs}} = 10$ independent seeds per cell, we obtain the results in Fig. 3 and Table 2.

Table 2: Multi-seed PRNG study at von Neumann scale: across-run mean and standard deviation of $\hat{k}$ for each (RNG, N) cell, with $n_{\text{runs}} = 10$, $N_{\text{active}} = 20$, $N_{\text{inactive}} = 20$.

	Middle Square		LCG		Mersenne Twister	
N	$\bar{k}$	σ_{seeds}	$\bar{k}$	σ_{seeds}	$\bar{k}$	σ_{seeds}
50	0.9524	0.3198	1.0860	0.0655	1.0970	0.0481
100	0.9603	0.3246	1.0837	0.0280	1.1042	0.0266
150	0.9723	0.3268	1.0888	0.0216	1.0929	0.0321
200	0.9762	0.3267	1.1020	0.0219	1.0871	0.0344

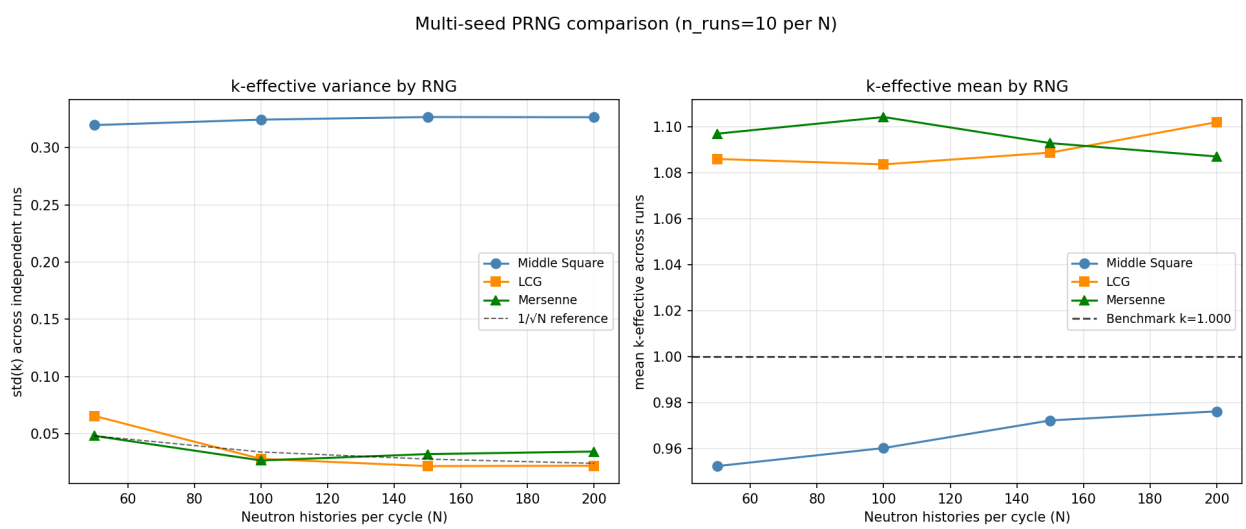

Figure 3: Multi-seed PRNG study at von Neumann scale ($n_{\text{runs}} = 10$ per N). **Left:** across-run standard deviation σ_{seeds} of $\hat{k}$ versus N. Middle-square's σ_{seeds} is approximately constant from $N = 50$ to $N = 200$; LCG and Mersenne both decrease, broadly tracking the $1/\sqrt{N}$ reference (dashed black). **Right:** across-run mean $\bar{k}$ versus N. LCG and Mersenne agree within statistical uncertainty and sit near the simulation's "true" value of $\hat{k} \approx 1.10$ established by the validation run; middle-square sits systematically $\sim$10% below at every N.

Three observations follow.

(1) LCG and Mersenne Twister are statistically indistinguishable. At every N tested, their mean estimates $\bar{k}$ agree to within $1\sigma_{\text{seeds}}$, and their standard deviations bracket the same $1/\sqrt{N}$ reference curve. At this scale and with this number of RNG calls per simulation ($\sim 10^4$–10^5, well

below either generator's period), the two modern generators produce statistically interchangeable results.

(2) Middle-square is biased. Its mean $\bar{k} \approx 0.96$ sits roughly 0.13 below the $\bar{k} \approx 1.09$ of the modern generators, a $\sim 10\%$ shift, persistent across all four values of N and across all 10 independent seeds per N. The bias is therefore not attributable to one unlucky seed. The mechanism is harder to pin down without additional analysis — the relevant feature would be some non-uniformity or autocorrelation in the $2{,}000$-block as a sample of $[0, 1]$ that systematically biases path-length sampling, isotope sampling, or scatter direction sampling — but the empirical fact of the bias is robust.

(3) Middle-square's σ_{seeds} does not decrease with N. This is the strongest finding. In a properly mixed Monte Carlo, where the N neutron histories per cycle are approximately i.i.d., the central limit theorem predicts $\sigma_{\text{seeds}} \propto 1/\sqrt{N}$. LCG and Mersenne obey this prediction. Middle-square does not: its σ_{seeds} goes from 0.320 at $N = 50$ to 0.327 at $N = 200$ — a 2% *increase*, against an expected factor-of-2 decrease. Adding neutrons to the simulation does not reduce the variance of the final estimate.

The mechanism for (3) is concrete. At $N = 100$ histories per cycle, with $\sim$5–10 collisions per neutron and $\sim$4–8 RNG calls per collision, a single cycle consumes several thousand random numbers — twice the block size of $2{,}000$ or more. Each cycle sees the same $2{,}000$ numbers in the same order, multiple times. Adding more neutrons within a cycle does not add new randomness; it merely re-uses the same cyclic structure more densely. The simulation's effective stochasticity is bounded above by the block diversity, not the neutron count. Run-to-run variance becomes dominated by which $2{,}000$-block was generated from which seed, and seeds vary across runs but the underlying combinatorial constraint does not vanish with N.

3.4 PRNG Comparison at Larger N

To verify that the von Neumann scale findings persist outside the historical operating point, we repeated the multi-seed study at $N \in \{200, 500, 1000\}$ with $n_{\text{runs}} = 10$ (Fig. 4, full numbers in Appendix D). At larger N, the simulation cycles through the middle-square block many tens of times per run, so within-run averaging over the block becomes possible and middle-square's σ_{seeds} does begin to decrease — but it tracks no clean scaling, dropping from 0.327 at $N = 200$ to 0.034 at $N = 500$ and back up to 0.089 at $N = 1{,}000$. Both modern generators track the $1/\sqrt{N}$ reference within sampling uncertainty (the apparent steeper decrease for LCG is within the $\sim 24\%$ relative uncertainty on σ_{seeds} at $n_{\text{runs}} = 10$). At $N = 1{,}000$, middle-square's σ_{seeds} is roughly an order of magnitude larger than either modern generator's (specifically $\sigma_{\text{ms}} \approx 0.089$ vs. $\sigma_{\text{LCG}} \approx 0.0035$, a factor of $\sim$25), and the bias of its mean ($\bar{k} \approx 1.04$ vs. $\bar{k} \approx 1.09$ for the modern generators) persists.

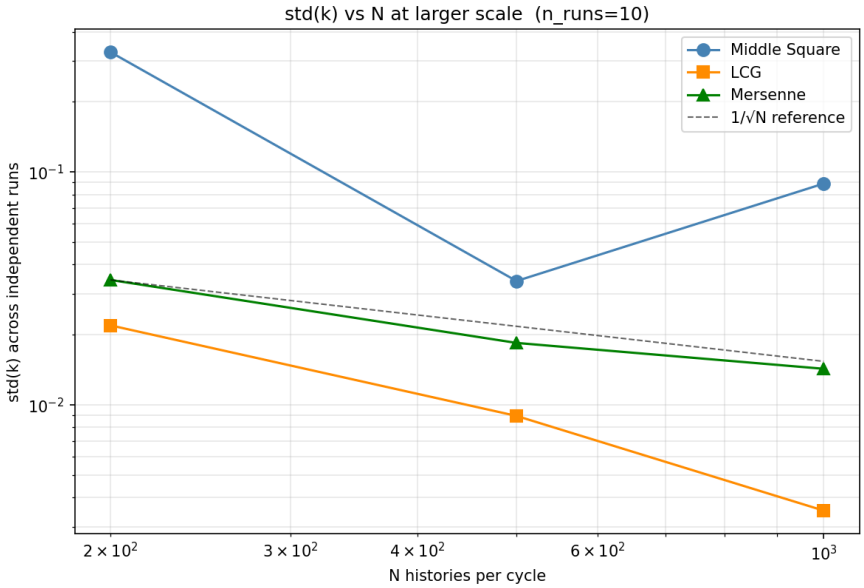

Figure 4: Multi-seed PRNG study at larger N on log-log axes ($n_{\text{runs}} = 10$ per cell). LCG and Mersenne both decrease with N, broadly tracking the $1/\sqrt{N}$ reference (both decrease within sampling uncertainty). Middle-square does begin to decrease at large N but follows no clean scaling: the curve is non-monotone and the values remain roughly an order of magnitude above the modern generators across all N tested.

4 Discussion

4.1 Tying Back to the Project Objectives

On Q1 (statistical convergence with N): both modern PRNGs broadly track the $1/\sqrt{N}$ scaling of σ_{seeds} with N within sampling uncertainty, with sharper agreement at larger N where statistics are more favorable. The prediction holds for the modern generators, where the i.i.d. assumption underlying CLT is approximately satisfied, and fails for middle-square, where block reuse violates that assumption (see Q2).

On Q2 (does PRNG quality matter?): yes, decisively. Within this study, middle-square exhibits a $\sim$10% systematic bias at the von Neumann scale ($N \leq 200$); at larger N the bias is smaller and more variable, statistically consistent with the modern generators at $N = 500$ and rebounding to $\sim$5% at $N = 1{,}000$. Its run-to-run standard deviation σ_{seeds} is approximately an order of magnitude above either modern generator's at $N = 200$ and $N = 1{,}000$, dipping toward parity at $N = 500$, and at the historical von Neumann operating point it is flat with N — a direct empirical violation of the central-limit prediction $1/\sqrt{N}$. At larger N (where each simulation cycles through the 2,000-block many tens of times) the middle-square σ_{seeds} does decrease, but follows no clean scaling. The mechanism — cyclic reuse of a fixed 2,000-number block, with the simulation's effective period bounded above by the block size rather than the recurrence cycle — is intrinsic to operating the middle-square at any scale where the per-cycle RNG demand exceeds the available cycle length of the seed, not to the mathematics of the recurrence as such. This demonstrates that, at the historical operating point, simulation precision is limited by block diversity rather than N

— a constraint the original 1948 calculations could not have observed because the conceptual and computational tools to compare against modern generators did not exist. More generally, this is a constraint on any Monte Carlo simulation: when per-iteration RNG demand exceeds the effective period of the generator, additional sample size N stops buying convergence.

4.2 The Simulation as a Platform

A useful way to read the results above is that the simulation is sufficiently controlled to be *used as a platform* for asking the RNG-quality question, even though it has a known 10% systematic from inelastic kinematics. All three PRNGs run through the same pipeline; their outputs are compared as relative quantities; and the inelastic systematic affects all three equally and is therefore irrelevant for cross-RNG comparisons. The simulation does not need to match the benchmark exactly to answer the proposal's central question — it needs to be a reproducible, internally consistent procedure that responds to RNG choice. It is.

4.3 Limitations

Inelastic kinematics. The most significant limitation is the elastic-kinematic treatment of inelastic scattering. Real inelastic events deposit $\sim$0.5–2 MeV per scatter into nuclear excited states; we assume effectively zero energy loss. This is the dominant mechanism behind our 10% overshoot of benchmark. A proper treatment requires sampling exit energies from the nuclear-level structure or from a Weisskopf-evaporation spectrum at an effective nuclear temperature ($\sim$0.5 MeV for fast neutrons on U-235); both are standard in production codes like MCNP and OpenMC.

Diffusion-theory comparison. The flux-profile overlay in §3.2 uses the simplest analytic mode of the diffusion equation with the Milne extrapolation length $0.7104\lambda_{\text{tr}}$. This is a leading-order approximation to the transport solution: it neglects the angular-flux structure that a full P_n or S_n treatment would resolve, which is the likely source of the $\sim$0.20 peak-normalized gap observed in the bulk of the sphere (§3.2). A more rigorous benchmark would use a converged transport solution from a production code (e.g. MCNP) rather than the diffusion mode used here; the present comparison is sufficient to verify that the burn-in cycles produce the expected center-peaked fundamental-mode shape, but not to quantify the magnitude of higher-order corrections.

Statistics at the largest N. Even with $n_{\text{runs}} = 10$ at $N = 1000$, the standard error on the across-seed standard deviation is non-negligible ($\sim 1/\sqrt{2(n_{\text{runs}} - 1)} \approx 24\%$ relative). The qualitative picture (middle-square fails to converge as $1/\sqrt{N}$) is robust to this uncertainty, but quantitative fits to the convergence rate would require larger n_{runs}.

Mechanism for middle-square bias not isolated. We characterize the bias and variance behavior empirically but do not identify the specific distributional or correlational property of the 2,000-block that produces a $\sim$10%-low $\hat{k}$. Candidate mechanisms include short-range autocorrelation in the block biasing path-length sampling (small successive draws giving systematically longer mean free paths), or non-uniformity in the block's marginal distribution biasing direction sampling. A related concern is that we did not verify the recurrence cycle length of every seed used in the multi-seed sweep: our implementation only fails fast if the recurrence reaches zero within `block_size` iterations, but a seed with recurrence cycle $C < 2{,}000$ would silently produce a block containing only C distinct values cycled through $\lceil 2{,}000/C \rceil$ times. Across seeds, variation in C would inflate σ_{seeds} in a way our protocol does not isolate. Disentangling all of these would

require diagnostic runs with single-channel analyses (path-length-only, direction-only) or controlled perturbations of the block (sub-sampling, shuffling, fixed-cycle-length filtering).

4.4 Future Directions

The cleanest follow-up is a proper inelastic energy-loss model, which we expect to bring the validation $\hat{k}$ into agreement with the benchmark to within a percent. A second is a sensitivity sweep over the sphere radius — a natural extension demonstrating the simulation's behavior near criticality. Third, on the RNG side, the bias mechanism characterized in Section 3.3 could be probed directly through sub-channel analyses or by varying the block size to test whether bias scales with effective period. Finally, while the bare HEU sphere is a clean test case, the same methodology — multi-seed sweeps comparing modern and historical RNGs — could be applied to reflected geometries or multi-region assemblies of greater historical or engineering interest.

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A Full Boltzmann Transport Equation

The intuitive form Eq. (1) suppresses the spatial, energy, and angular dependencies of every quantity for readability. The full form, with all dependencies explicit, is:

$$\begin{aligned}\frac{1}{v(E)}\frac{\partial\psi(\mathbf{r},E,\mathbf{\Omega},t)}{\partial t} &= Q(\mathbf{r},E,\mathbf{\Omega},t) + \int_0^\infty dE' \int_{S^2} d\Omega'\, \Sigma_s(\mathbf{r},E'\to E,\mathbf{\Omega}'\to\mathbf{\Omega})\,\psi(\mathbf{r},E',\mathbf{\Omega}',t)\\ &\quad + \frac{\chi(E)}{4\pi}\int_0^\infty dE' \int_{S^2} d\Omega'\,\nu(E')\,\Sigma_f(\mathbf{r},E')\,\psi(\mathbf{r},E',\mathbf{\Omega}',t)\\ &\quad - \mathbf{\Omega}\cdot\nabla_{\mathbf{r}}\psi(\mathbf{r},E,\mathbf{\Omega},t) - \Sigma_t(\mathbf{r},E)\,\psi(\mathbf{r},E,\mathbf{\Omega},t). \end{aligned} \tag{8}$$

Throughout, primed variables (E', $\mathbf{\Omega}'$) are dummy integration variables ranging over source phase space; unprimed variables (E, $\mathbf{\Omega}$) are the destination point at which the equation is evaluated. The differential scattering cross section $\Sigma_s(\mathbf{r}, E' \to E, \mathbf{\Omega}' \to \mathbf{\Omega})$ encodes the probability per unit length per unit final energy per unit final solid angle that a neutron at energy E' and direction $\mathbf{\Omega}'$ scatters into energy E and direction $\mathbf{\Omega}$. In the fission term, $\nu(E')$ is the (energy-dependent) mean number of secondary neutrons released per fission of a target nucleus by an incoming neutron of energy E', and $\chi(E)$ is the fission spectrum — the energy distribution of the secondaries, modeled here by the Watt spectrum (Appendix B). Note that $\chi(E)/4\pi$ factors outside the fission integral because fission daughters' energy and direction depend only on the destination $(E, \mathbf{\Omega})$, not the parent's $(E', \mathbf{\Omega}')$. The scatter kernel does not factor: elastic kinematics tie outgoing energy to scattering angle.

B Watt Spectrum Sampling

Fission neutrons are born with energies distributed according to the Watt spectrum

$$p(E) \propto \exp(-E/a)\,\sinh\!\left(\sqrt{b\,E}\right), \tag{9}$$

with U-235 parameters $a = 0.988$ MeV, $b = 2.249$ MeV^{-1} [2]. We sample E via the Maxwell-with-correction method of Brown [4], which is the algorithm used in the production codes MCNP and OpenMC:

1. Sample w from $\mathrm{Maxwell}(a)$ via

$$w = -a\left[\ln\xi_1 + \ln\xi_2\cdot\cos^2\!\left(\tfrac{\pi}{2}\xi_3\right)\right], \quad \xi_1,\xi_2,\xi_3 \sim \mathrm{Uniform}(0,1). \tag{10}$$

2. Apply the Watt correction

$$E = w + \tfrac{1}{4}a^2 b + (2\xi_4 - 1)\sqrt{a^2 b w}, \quad \xi_4 \sim \mathrm{Uniform}(0,1). \tag{11}$$

This produces an exact draw from the Watt distribution: the algorithm samples w from a Maxwell distribution and applies a w-dependent shift designed to match the Watt density; the derivation is in Brown [4]. A sanity-check histogram of 10^5 samples against the analytical density gives mean $\langle E \rangle = 2.04$ MeV (analytical: 2.03 MeV) and follows the theoretical curve within sampling noise across the energy range $[0, 12]$ MeV (Fig. 5).

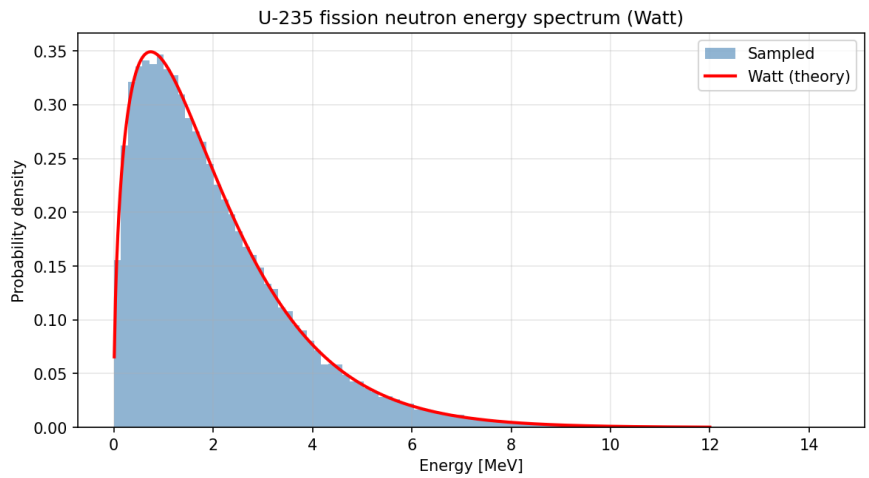

Figure 5: Sanity check of the Watt sampler: histogram of 10^5 sampled energies (blue) against the analytic Watt density (orange) for U-235 parameters $a = 0.988$ MeV, $b = 2.249$ MeV^{-1}. Sample mean $\langle E \rangle = 2.04$ MeV agrees with the analytical value 2.03 MeV.

C Sampling Formulas for the Neutron History

A single neutron history advances by repeated application of the following sampling steps; see Lux & Koblinger [2] for full derivations. At each collision, the joint distribution over (path length, isotope, reaction) is sampled sequentially via the chain rule of probability: each conditional draws on what has already been determined. Path length is sampled marginally (no conditioning), isotope conditional on a collision having occurred, and reaction conditional on the isotope.

Path length. The distance to the next collision is drawn from the exponential distribution via inverse-CDF. If $F(d) = 1 - e^{-\Sigma_t d}$ then $F^{-1}(\xi) = -\ln(1-\xi)/\Sigma_t$; since $1 - \xi \overset{d}{=} \xi$ for $\xi \sim \mathrm{Uniform}(0, 1)$,

$$d = -\frac{\ln \xi}{\Sigma_t}. \tag{12}$$

Boundary check. The candidate next position is $\mathbf{r}' = \mathbf{r} + d\,\mathbf{\Omega}$. If $|\mathbf{r}'| > R$, the neutron escapes the sphere and is removed. The exact escape distance is the positive root of $|\mathbf{r} + t\mathbf{\Omega}|^2 = R^2$.

Isotope selection. Given that a collision occurred, the target isotope is sampled proportional to its contribution to Σ_t:

$$P(\text{hit isotope } i) = \frac{n_i\, \sigma_t^{(i)}(E)}{\Sigma_t(E)}. \tag{13}$$

Reaction selection. Given isotope i, the reaction channel is sampled from the partial cross sections $\sigma_x^{(i)}(E)$ for $x \in \{\text{fission}, \text{elastic}, \text{inelastic}, \text{capture}\}$:

$$P(\text{reaction } x \mid i) = \frac{\sigma_x^{(i)}(E)}{\sigma_t^{(i)}(E)}. \tag{14}$$

Elastic and inelastic are combined into a single "scatter" branch (the neutron survives in both cases) and the elastic kinematic formula is applied to the surviving neutron.

Elastic scatter kinematics. Isotropic in the center-of-mass frame: $\mu_0 = \cos\theta_{\text{CM}} \sim \text{Uniform}(-1, 1)$, $\phi \sim \text{Uniform}(0, 2\pi)$. The choice of $\mu = \cos\theta$ (rather than θ itself) as the uniformly distributed variable comes from the spherical surface element: $dA = \sin\theta\, d\theta\, d\phi = -d\mu\, d\phi$. In (μ, ϕ) coordinates the sphere's surface measure is uniform on the rectangle $[-1, 1] \times [0, 2\pi]$, so uniform sampling in these coordinates produces uniform-on-sphere directions automatically — no Jacobian correction needed. The same μ-uniform construction is used for initial neutron birth and for fission daughters. In the lab frame the post-scatter energy is

$$E' = E \cdot \frac{1 + \alpha + (1 - \alpha)\mu_0}{2}, \quad \alpha = \left(\frac{A - 1}{A + 1}\right)^2, \tag{15}$$

where A is the mass number of the target nucleus (234, 235, or 238 for our isotopes). The factor α is the minimum fraction of energy retained: E'/E ranges linearly between α (at $\mu_0 = -1$, full backscatter) and 1 (at $\mu_0 = +1$, no deflection). For U-235 ($A = 235$), $\alpha \approx 0.983$, so the maximum per-scatter energy loss is $\sim 1.7\%$ — the quantitative basis for the claim in §3 that neutrons in our simulation stay near $\sim$2 MeV throughout their lives. The new direction $\mathbf{\Omega}'$ is obtained by rotating $\mathbf{\Omega}$ by (μ_0, ϕ) via the standard CM-to-lab rotation [2]. In the multi-speed model implemented here, E' is then used to recompute cross sections at the new energy before the next collision.

Fission yield. On average $\bar{\nu} = 2.58$ neutrons are released per fission for U-235 at fast energies [2] (held constant rather than tabulated as $\bar{\nu}(E)$, consistent with the simulation's inelastic-kinematics simplification; see §2 for discussion). Since ν must be integer, we sample

$$\nu = \lfloor \bar{\nu} \rfloor + \text{Bernoulli}(\bar{\nu} - \lfloor \bar{\nu} \rfloor) = 2 + \text{Bernoulli}(0.58), \tag{16}$$

preserving the correct mean. Each fission neutron is born at the parent's collision site with isotropic direction (resampled) and energy drawn from the Watt spectrum (Appendix B).

D Multi-Seed PRNG Study: Full Numerical Results

Table 3: Multi-seed PRNG study at von Neumann scale. $\bar{k}$ is the mean of $\hat{k}$ across $n_{\text{runs}} = 10$ independent seeds; σ_{seeds} is the standard deviation across runs. $N_{\text{active}} = 20$, $N_{\text{inactive}} = 20$.

	Middle Square		LCG		Mersenne Twister	
N	$\bar{k}$	σ_{seeds}	$\bar{k}$	σ_{seeds}	$\bar{k}$	σ_{seeds}
50	0.9524	0.3198	1.0860	0.0655	1.0970	0.0481
100	0.9603	0.3246	1.0837	0.0280	1.1042	0.0266
150	0.9723	0.3268	1.0888	0.0216	1.0929	0.0321
200	0.9762	0.3267	1.1020	0.0219	1.0871	0.0344

Table 4: Multi-seed PRNG study at larger N, same protocol with $n_{\text{runs}} = 10$, $N_{\text{active}} = 20$, $N_{\text{inactive}} = 20$.

	Middle Square		LCG		Mersenne Twister	
N	$\bar{k}$	σ_{seeds}	$\bar{k}$	σ_{seeds}	$\bar{k}$	σ_{seeds}
200	0.9762	0.3267	1.1020	0.0219	1.0871	0.0344
500	1.0690	0.0340	1.0887	0.0090	1.0921	0.0184
1000	1.0400	0.0890	1.0883	0.0035	1.0962	0.0143

Code and Data Availability

The simulation code (`simulate.py` for the transport simulation, PRNG implementations, multi-seed analysis, and plotting; `cross_sections.py` for the data loader) is available at https://github.com/jakobrees/neutron-transport-prng. The relevant ENDF/B-VIII.0 [8] evaluations for U-234, U-235, and U-238 (reactions MT1, MT2, MT18, MT102) are bundled with the repository as gzipped JSON files; the original tabulated data is publicly available from the National Nuclear Data Center at Brookhaven (https://www.nndc.bnl.gov/endf/).